Lecture 4: Using Basic Integration Formula

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Perform basic integration by using basic integration formula.
- ☐ Apply suitable substitution in computing integrals.

Introduction

Recall that if $F(x)$ is a differentiable function of x with $F(x) = f'(x)$, then

$$\int f(x)\,dx = F(x) + C.$$

Also recall that if we let $u = g(x)$, then

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du.$$

Basic Integration Formulas

1. $\int k\,dx = kx + C$ (any number k)

2. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

3. $\int \frac{dx}{x} = \ln|x| + C$

4. $\int e^x\,dx = e^x + C$

5. $\int a^x\,dx = \frac{a^x}{\ln a} + C \quad (a > 0,\ a \neq 1)$

6. $\int \sin x\,dx = -\cos x + C$

7. $\int \cos x\,dx = \sin x + C$

8. $\int \sec^2 x\,dx = \tan x + C$

9. $\int \csc^2 x\,dx = -\cot x + C$

10. $\int \sec x \tan x\,dx = \sec x + C$

11. $\int \csc x \cot x\,dx = -\csc x + C$

12. $\int \tan x\,dx = \ln|\sec x| + C$

13. $\int \cot x\,dx = \ln|\sin x| + C$

14. $\int \sec x\,dx = \ln|\sec x + \tan x| + C$

15. $\int \csc x\,dx = -\ln|\csc x + \cot x| + C$

16. $\int \sinh x\,dx = \cosh x + C$

17. $\int \cosh x\,dx = \sinh x + C$

18. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C$

19. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$

20. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\operatorname{arcsec}\left|\frac{x}{a}\right| + C$

21. $\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C \quad (a > 0)$

22. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C \quad (x > a > 0)$

Examples

$$\int_3^5 \frac{2x-3}{\sqrt{x^2-3x+1}}\,dx$$

$$\int \frac{dx}{\sqrt{8x-x^2}}$$

Examples

$$\int (\cos x \sin 2x + \sin x \cos 2x)\, dx$$

$$\int_0^{\frac{\pi}{4}} \frac{dx}{1 - \sin x}$$

Examples

$$\int \frac{3x^2 - 7x}{3x + 2}\, dx$$

$$\int \frac{3x + 2}{\sqrt{1 - x^2}}\, dx$$

Examples

$$\int \frac{dx}{(1+\sqrt{x})^3}$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^3 \cos x \, dx$$

Examples

Evaluate $\int_0^1 \frac{x^3}{(4+x^2)^{\frac{5}{2}}}\,dx$ by using the substitution

- $x = 2\tan\theta$

- $x = 2\sinh\theta$

Examples

Evaluate $\int_0^1 \frac{x^3}{(4+x^2)^{\frac{5}{2}}}\,dx$ by using the substitution

- $u^2 = 4 + x^2$

- $u = 4 + x^2$